

σ -Algebra.

Def. Let X be a set.

A collection $\mathcal{M} \subseteq \mathcal{P}(X)$ is called a σ -algebra in X if:

- 1) $X \in \mathcal{M}$.
- 2) If $A \in \mathcal{M}$, then $A^c \in \mathcal{M}$.
- 3) If $A = \bigcup_{n=1}^{\infty} A_n$ and if $A_n \in \mathcal{M}$ for $n \in \mathbb{N}$, then $A \in \mathcal{M}$.

For the such a σ -algebra $\mathcal{M}$, $(X, \mathcal{M})$ is called a measurable space.

And, an element of $\mathcal{M}$ is called a measurable set.

Def. Let $(X, \mathcal{M})$ be a measurable space and

$(Y, \mathcal{U})$ be a topological space.

A function $f: X \rightarrow Y$ is called a measurable if:

for any open set $U \subseteq Y$, $f^{-1}[U] \subseteq X$ is a measurable set.

That is, $U \in \mathcal{U} \Rightarrow f^{-1}[U] \in \mathcal{M}$.

Thm. Let Y and Z be topological spaces, $g: Y \rightarrow Z$ be a Continuous.

If X is a measurable space and $f: X \rightarrow Y$ is a measurable, then $g \circ f$ is a measurable.

Proof. Given an open set $U \subseteq Z$, $g^{-1}[U] \subseteq Y$ is open, so $(g \circ f)^{-1}[U] = f^{-1}[g^{-1}[U]]$ is measurable. $\square$

Borel set.

Thm. Let X be a set.

For any collection $\mathcal{F} \subseteq \mathcal{P}(X)$ of X , there exists a smallest σ -algebra $\mathcal{M}^*$ containing $\mathcal{F}$ in X .

Proof. Let $\mathcal{O}$ be the family of all σ -algebras which contain $\mathcal{F}$.

That is, $\mathcal{O} = \{ \mathcal{M} \subseteq \mathcal{P}(X) \mid \mathcal{M} \text{ is } \sigma\text{-algebra in } X, \text{ and } \mathcal{F} \subseteq \mathcal{M} \}$.

Since $\mathcal{P}(X)$ is a σ -algebra in X itself and contains $\mathcal{F}$, therefore $\mathcal{O}$ is non-empty set. Define

$$\mathcal{M}^* = \bigcap_{\mathcal{M} \in \mathcal{O}} \mathcal{M}$$

clearly $\mathcal{F} \subseteq \mathcal{M}^*$, and $\mathcal{M}^* \subseteq \mathcal{M}$ for any $\mathcal{M} \in \mathcal{O}$.

Now, enough to show that $\mathcal{M}^*$ is a σ -algebra in X .

- 1) Since $X \in \mathcal{M}$ for all $\mathcal{M} \in \mathcal{O}$, $X \in \mathcal{M}^*$.
- 2) If $A \in \mathcal{M}^*$, then $A^c \in \mathcal{M}$ for all $\mathcal{M} \in \mathcal{O}$, thus $A^c \in \mathcal{M}^*$.
- 3) Assume $A_n \in \mathcal{M}^*$ ($n \in \mathbb{N}$). Then, for any $\mathcal{M} \in \mathcal{O}$, $A_n \in \mathcal{M}$ for all $n \in \mathbb{N}$, and so that $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$. Thus, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}^*$. $\square$

Def. Let $(X, \mathcal{C})$ be a topological space.

The σ -algebra $\mathcal{B}$ in X generated by $\mathcal{C}$ is called the Borel σ -algebra, and its element is called a Borel set.

Moreover, a continuous map $f: (X, \mathcal{C}) \rightarrow (Y, \mathcal{C}')$ is measurable with respects to $(X, \mathcal{B})$ where $\mathcal{B}$ is Borel σ -algebra generated by $\mathcal{C}$, and such a f is called a Borel function.

Note: In Topology, F_σ is a set of all countable union of closed set.
 G_δ is a set of all countable intersection of opens.

Thm. Let $\mathcal{M}$ be a σ -algebra in X , and Y be a topological sp.
Let $f: X \rightarrow Y$ be a map. Then, the followings hold:

- a). $\mathcal{O} = \{E \subseteq Y \mid f^{-1}[E] \in \mathcal{M}\}$ is a σ -algebra in Y .
- b). If f is measurable and E is a Borel set in Y , then $f^{-1}[E] \in \mathcal{M}$.
- c). If $Y = [-\infty, \infty]$ and $f^{-1}[(a, \infty]) \in \mathcal{M}$ for any $a \in \mathbb{R}$, then f is measurable.
- d).